

CVS · DAY 6 · FLASHCARDS

USMLE Step 1 · June Cohort · 52 cards

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bakesiga.github.io/usmle-dashboard*Cover the answer · attempt aloud · check · re-test the misses tomorrow***HF CLINICAL PRESENTATION****6 CARDS****1 Classic constellation suggesting decompensated left HF on bedside exam?**

Orthopnea + paroxysmal nocturnal dyspnea + bibasilar crackles + **S3 gallop** + JVD + lower-extremity pitting edema, often with an apical impulse displaced **inferolaterally** from LV dilation.

2 Most common cause of right-sided heart failure?

Left-sided heart failure. Second most common is chronic pulmonary disease (cor pulmonale).

3 Hemosiderin-laden macrophages in alveolar spaces — what do they indicate and what stain confirms?

Chronic left ventricular failure ("**heart failure cells**"). RBCs leak across pulmonary capillaries → ingested by alveolar macrophages → iron stored as hemosiderin → **Prussian blue positive**.

4 Why might a patient have markedly elevated filling pressures yet minimal peripheral edema?

The **lymphatic system has ~10–20× reserve capacity** for interstitial fluid absorption. Clinical edema only appears once the lymphatics are saturated.

5 Histology of acute pulmonary edema?

Pink, acellular intra-alveolar transudate filling the alveoli, ± distended alveolar capillaries.

6 Bedside signs distinguishing right HF / cor pulmonale from left HF?

Right-sided: **JVD, hepatomegaly, hepatojugular reflux, lower-extremity edema, ascites**, Kussmaul sign. Left-sided: orthopnea, PND, crackles, S3, displaced PMI.

S3 VS S4**4 CARDS****7 S3: when in the cycle does it fall, what loading condition does it imply, and what causes it mechanically?**

Just **after S2**, in early diastole. Implies **volume overload** / floppy non-compliant LV (HFrEF, severe MR, severe AR, high-output states). Caused by rapid early-diastolic ventricular filling decelerating against a dilated LV.

8 S4: when in the cycle does it fall, what loading condition does it imply, and what causes it mechanically?

Just **before S1**, late diastole. Implies **pressure overload** / stiff LV (chronic HTN with concentric LVH, HCM, restrictive cardiomyopathy, acute MI). Caused by the atrial kick contracting blood into a non-compliant ventricular wall.

9 In which physiologic population can an S3 be normal?

Healthy individuals **under age 40**, athletes, and pregnant women — a compliant LV can produce a benign S3.

10 Best position and stethoscope technique to hear S3 and S4?

Left lateral decubitus position, with the **bell** of the stethoscope at the cardiac apex.

HFREF VS HFPEF

5 CARDS

11 Ejection fraction cut-offs for HFrEF, HFmrEF, and HFpEF?

HFrEF $\leq 40\%$; HFmrEF 41–49%; HFpEF $\geq 50\%$.

12 Dominant driver and LV geometry of HFpEF?

Dominant driver: **chronic hypertension** → **concentric LV hypertrophy** with a non-dilated chamber. Reflects increased afterload causing diastolic dysfunction.

13 How does the pressure-volume loop change in HFpEF compared with normal?

The **end-diastolic pressure-volume relationship (EDPVR)** shifts **up and to the left** — the ventricle requires much higher diastolic pressures to achieve a given filling volume because of stiffness.

14 Mechanism of HFrEF on the PV loop?

ESPVR slope decreases ($\downarrow$ contractility) and the whole loop shifts rightward ($\uparrow$ end-diastolic and end-systolic volumes). Stroke volume falls; ejection fraction falls.

15 Reversible causes of dilated cardiomyopathy?

Alcoholic DCM (with abstinence), peripartum, tachycardia-induced, stress (Takotsubo), hyperthyroid, cocaine, thiamine deficiency.

ACC/AHA STAGE & NYHA CLASS

4 CARDS

16 Define ACC/AHA stages A through D.

A = at risk (HTN, DM, CAD), no structural disease. **B** = structural disease but asymptomatic. **C** = symptomatic HF. **D** = refractory HF requiring advanced therapy.

17 At which ACC/AHA stage should ACE-i and β -blocker be started in HFrEF — and why?

Stage B — asymptomatic structural disease. By Stage B the neurohormonal cascade (sympathetic, RAAS, ANP) is already activated, so blocking it early slows progression to symptomatic HF.

18 Define NYHA class I through IV.

I = asymptomatic with ordinary activity. **II** = symptoms with ordinary activity. **III** = symptoms with less than ordinary activity. **IV** = symptoms at rest.

19 Key difference between ACC/AHA staging and NYHA classification?

ACC/AHA stages describe **structural progression** (one-way: A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D). NYHA describes **current functional status** and can move both directions (e.g., III $\rightarrow$ II with treatment).

ADHF CHEST X-RAY

3 CARDS

20 List the chest X-ray findings of acute decompensated heart failure.

Bilateral patchy **airspace opacities** (alveolar edema); **Kerley B lines**; **bat-wing perihilar opacification**; thickened interlobar fissures; blunted costophrenic angles (effusions); **cardiomegaly**; cephalization (upper-lobe vessel diversion).

21 What are Kerley B lines and what do they represent?

Short, horizontal lines ≤ 2 cm at the lung bases, perpendicular to the pleural surface. They represent **thickened interlobular septa from interstitial pulmonary edema**.

22 How do the cardiac and vascular function curves shift in decompensated HFrEF?

Cardiac curve shifts **down and right** ($\downarrow$ contractility); vascular curve shifts **right** (volume retention raises mean systemic pressure). New intersection: **elevated right atrial pressure with maintained or slightly reduced cardiac output**.

BNP & NATRIURETIC PEPTIDES

4 CARDS

23 What cells release BNP, and what is the trigger?

Ventricular cardiomyocytes release BNP in response to **ventricular wall stress** (Laplace: $T = Pr / 2h$ — wall tension rises with chamber pressure and radius, falls with wall thickness).

24 Downstream signaling of ANP and BNP?

Both activate **guanylate cyclase** $\rightarrow \uparrow$ cGMP $\rightarrow$ natriuresis + diuresis + arterial and venous vasodilation + $\downarrow$ aldosterone + $\downarrow$ renin.

25 What is the most useful clinical interpretation of a normal BNP in a dyspneic patient?

A normal BNP has a **high negative predictive value** — it effectively rules out acute decompensated heart failure as the cause of dyspnea.

26 Conditions besides HF that elevate BNP, and one condition that falsely lowers it?

Elevated: AFib, CKD, PE, sepsis, advanced age, female sex. **Falsely low: obesity** — adipose tissue expresses neprilysin which clears BNP.

NEUROHORMONAL CASCADE

3 CARDS

27 Trace the neurohormonal vicious cycle of decompensated HF in sequence.

↓ CO → ↓ baroreceptor stretch + ↓ renal perfusion → **SNS** activation (↑ HR, ↑ contractility, vasoconstriction) + **RAAS** activation (AngII vasoconstriction + aldosterone-mediated Na/H₂O retention) → ↑ SVR (afterload) and ↑ preload → worsens HF → loop repeats. ANP/BNP try to counter (natriuresis + vasodilation) but are overwhelmed.

28 In asymptomatic Stage B HF, which neurohormones are already elevated?

All three: **angiotensin II, norepinephrine, and ANP**. This is the biological argument for starting ACE-i + β-blocker at Stage B before symptoms develop.

29 Where does ACE convert angiotensin I to angiotensin II, and how does this show up in HF?

In the **pulmonary vascular endothelium**. In HF, the pulmonary vein has measurably higher AngII concentration than the pulmonary artery — pulmonary circulation is the main site of conversion.

HFrEF FOUR-PILLAR REGIMEN

8 CARDS

30 Name the four mortality-reducing pillar classes for HFrEF.

(1) **β-blocker** (carvedilol, metoprolol succinate, bisoprolol); (2) **ACE-i / ARB / ARNi**; (3) **MRA** (spironolactone, eplerenone); (4) **SGLT2 inhibitor** (dapagliflozin, empagliflozin).

31 Which β-blockers are proven to reduce mortality in HFrEF?

Carvedilol, metoprolol succinate, and bisoprolol. Start low, go slow; do not initiate during acute decompensation.

32 How does sacubitril/valsartan (ARNi) act differently from an ACE inhibitor?

Sacubitril inhibits **neprilysin**, the enzyme that degrades natriuretic peptides → ↑ endogenous ANP/BNP activity. Paired with valsartan (ARB) to also block RAAS. **Superior to ACE-i** in stable HFrEF (PARADIGM-HF).

33 Why must ACE-i be washed out for 36 hours before starting sacubitril/valsartan?

Both block bradykinin breakdown — combining them risks severe **angioedema**.

34 Mechanism of the dry cough and angioedema seen with ACE inhibitors? Best alternative?

Bradykinin accumulation (ACE normally degrades bradykinin). Switch to an **ARB** — angiotensin receptor blockers do not affect bradykinin metabolism.

35 How do SGLT2 inhibitors benefit HF patients without diabetes?

They reduce HF hospitalization and cardiovascular death **regardless of diabetic status**. Exact mechanism is incompletely understood — proposed contributions include natriuresis, ↓ preload, improved myocardial energetics, and renal protection.

36 Role of loop diuretics in HFrEF — do they reduce mortality?

No. Loop diuretics (furosemide, torsemide, bumetanide) relieve congestion and improve symptoms but have no proven mortality benefit. Use the lowest effective dose.

37 Indications for ICD and CRT in HFrEF?

ICD: EF ≤ 35 % after ≥ 3 months of optimized GDMT (primary prevention of sudden cardiac death). **CRT (biventricular pacing):** EF ≤ 35 % + **QRS ≥ 150 ms** + **LBBB** morphology, with persistent NYHA II–IV symptoms.

HFpEF MANAGEMENT

2 CARDS

38 Which drug classes have the strongest evidence in HFpEF?

SGLT2 inhibitors (consistent reduction in HF hospitalization across the full EF spectrum); **MRA** (especially EF 45–55 %); plus aggressive **BP control** and management of comorbidities (AF, OSA, obesity, CKD). Loop diuretics for symptom control.

39 Why do β-blockers and ACE inhibitors offer less benefit in HFpEF than HFrEF?

HFpEF is driven by diastolic dysfunction and impaired ventricular relaxation, not by reduced contractility or a maximally activated neurohormonal cascade. β-blockade can even worsen exercise capacity by limiting chronotropy.

RIGHT HF · COR PULMONALE · CARDIOTOXIC CMP

4 CARDS

40 Pathophysiology of cor pulmonale in COPD?

Chronic alveolar hypoxia → **hypoxic pulmonary vasoconstriction** + vascular remodeling → pulmonary hypertension → **RV pressure overload** → RV hypertrophy/dilation → tricuspid regurgitation → right heart failure.

41 Cardiac ascites: SAAG and total-protein profile, and how this distinguishes it from cirrhotic ascites.

Both have **SAAG > 1.1 g/dL** (portal hypertension category). **Cardiac** ascites has **total protein > 2.5 g/dL** (sinusoidal congestion lets protein leak), while cirrhotic ascites has **total protein < 2.5 g/dL**.

42 Anthracycline (doxorubicin) cardiotoxicity — mechanism, time course, and preventive agent?

Mechanism: **free-radical mitochondrial damage** + **topoisomerase IIβ inhibition** in cardiomyocytes. **Dose-dependent and irreversible**; DCM can present years to decades after exposure. Prevented by **dexrazoxane**.

43 Why is alcoholic dilated cardiomyopathy potentially reversible while anthracycline DCM is not?

Alcoholic DCM reflects ongoing toxic insult to viable cardiomyocytes — abstinence allows recovery of contractility. Anthracyclines cause **permanent myocyte loss** via mitochondrial and DNA damage; the cells do not regenerate.

SHOCK — CATH PROFILES & FIRST-LINE TREATMENT

7 CARDS

44 Catheter profile of hypovolemic shock (CVP, PCWP, CI, SVR, SvO₂), and first-line treatment?

↓ CVP, ↓ PCWP, ↓ CI, ↑ SVR, ↓ SvO₂; cold extremities. Treatment: **rapid IV crystalloid** + source control (transfusion for hemorrhage, hemostasis, etc.).

45 Catheter profile of cardiogenic shock and first-line interventions in acute MI?

↑ CVP, ↑ PCWP, ↓ CI, ↑ SVR, ↓ SvO₂; cold. Treatment: **emergency revascularization (PCI)**, inotrope (**dobutamine**), mechanical support (IABP, Impella, ECMO) if refractory.

46 Catheter profile and pathophysiology of obstructive shock from massive PE?

↑ CVP, ↓ PCWP, ↓ CI, ↑ SVR, ↓ SvO₂. Pathway: PA clot → ↑ RV afterload → RV dilation + wall tension → RV ischemia → ↓ LV preload → cardiogenic collapse. Echo shows enlarged RV with septal flattening.

47 Treatment escalation for massive PE?

Anticoagulation in all; **systemic thrombolytics** (tPA) if hemodynamically compromised; catheter-directed therapy or **surgical embolectomy** for refractory or contraindication to lysis.

48 Catheter profile of warm septic shock and first-line treatment bundle?

↓ CVP, ↓ PCWP, ↑ CI, ↓ SVR, ↑ SvO₂ (impaired tissue O₂ extraction); warm extremities. Treatment: **early broad-spectrum antibiotics + IV crystalloid + norepinephrine** as first-line vasopressor.

49 Mechanism of anaphylactic shock and the single most important first-line intervention?

IgE-mediated mast cell degranulation → histamine + leukotrienes + tryptase → systemic vasodilation, capillary leak, bronchoconstriction. First-line: **IM epinephrine** (α₁ vasoconstriction + β₁ inotropy + β₂ bronchodilation + mast cell stabilization).

50 Why might PCWP appear paradoxically elevated in cardiac tamponade despite an under-filled LV?

The high pericardial pressure is **transmitted into all cardiac chambers**, so the catheter reads an elevated wedge pressure. Effective transmural filling pressure (and therefore preload) is actually low.

VASOPRESSOR / INOTROPE PAIRING

1 CARD

51 First-line vasopressor for each shock type: septic, anaphylactic, cardiogenic with adequate BP, hypovolemic?

Septic: norepinephrine (α₁ >> β₁). **Anaphylactic:** IM epinephrine. **Cardiogenic with adequate BP:** dobutamine (β₁ inotrope). **Hypovolemic:** volume first — pressors only as a bridge while volume is being replaced.

HIGH-OUTPUT HF

1 CARD

52 Hemodynamic signature of high-output heart failure from hyperthyroidism, and management of the Graves component?

↓ SVR + ↑ CO + wide pulse pressure + warm extremities + pitting edema, often with an S3 from volume overload. Graves treatment: **propranolol** (β-blockade + peripheral T4 → T3 conversion block) plus definitive therapy (**methimazole, radioactive iodine, or thyroidectomy**). Standard HF medications alone do not fix the underlying high-output state.